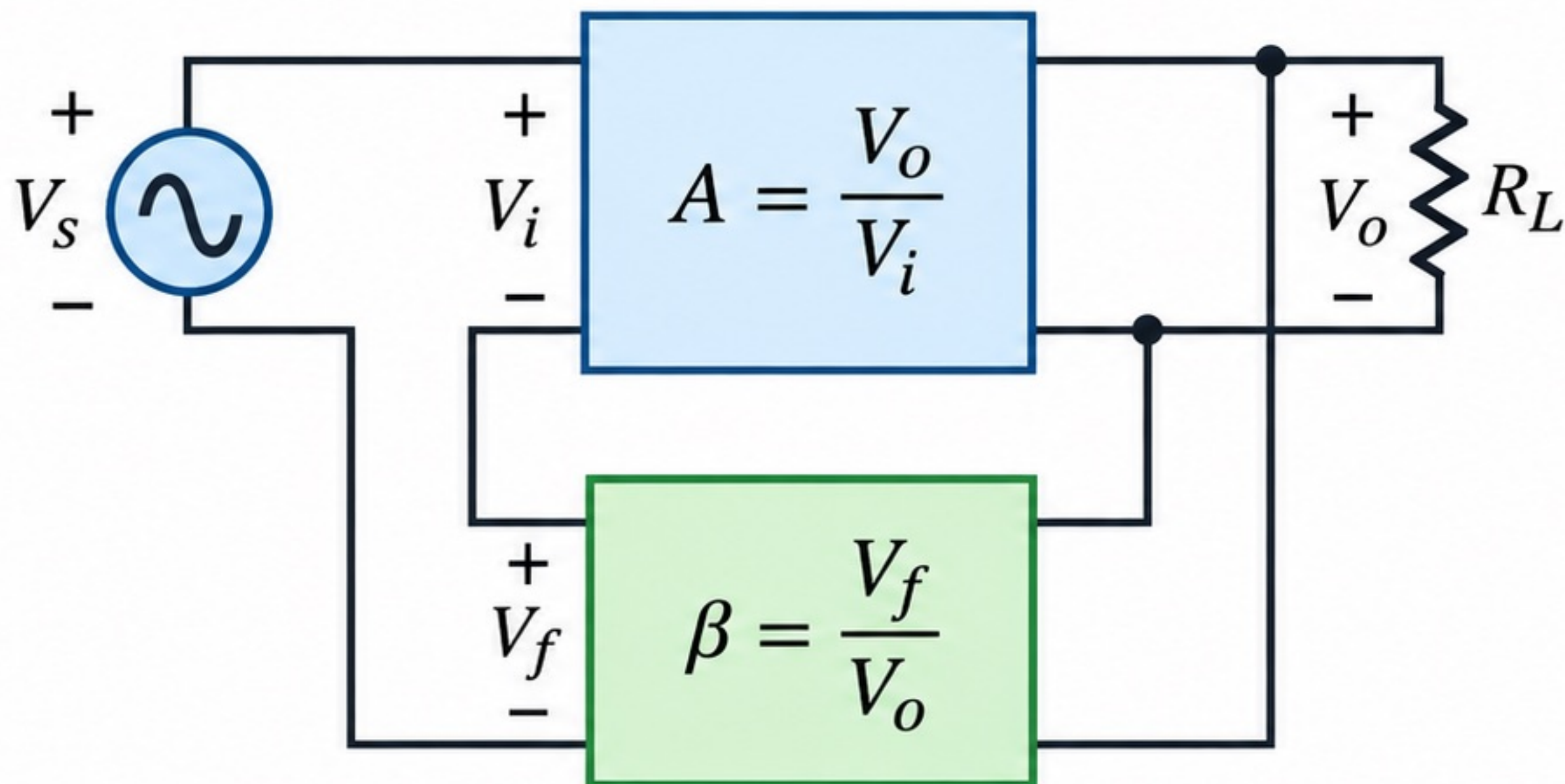


1. What Feedback Does

Negative feedback compares the output to the input and reduces error

ANALYZE
DESIGN
UNDERSTAND
APPLY



Key Ideas

- 1 The forward amplifier converts the error/input signal V_i into the output V_o .
- 2 The feedback network samples the output and returns $V_f = \beta V_o$.
- 3 For negative feedback, the effective input is $V_i = V_s - V_f$.
- 4 The returned signal opposes the source signal's effect, reducing error.
- 5 **Result:** lower raw gain, but better accuracy, robustness, and impedance control.



Feedback compares what we want (V_s) with what we got (through V_f).

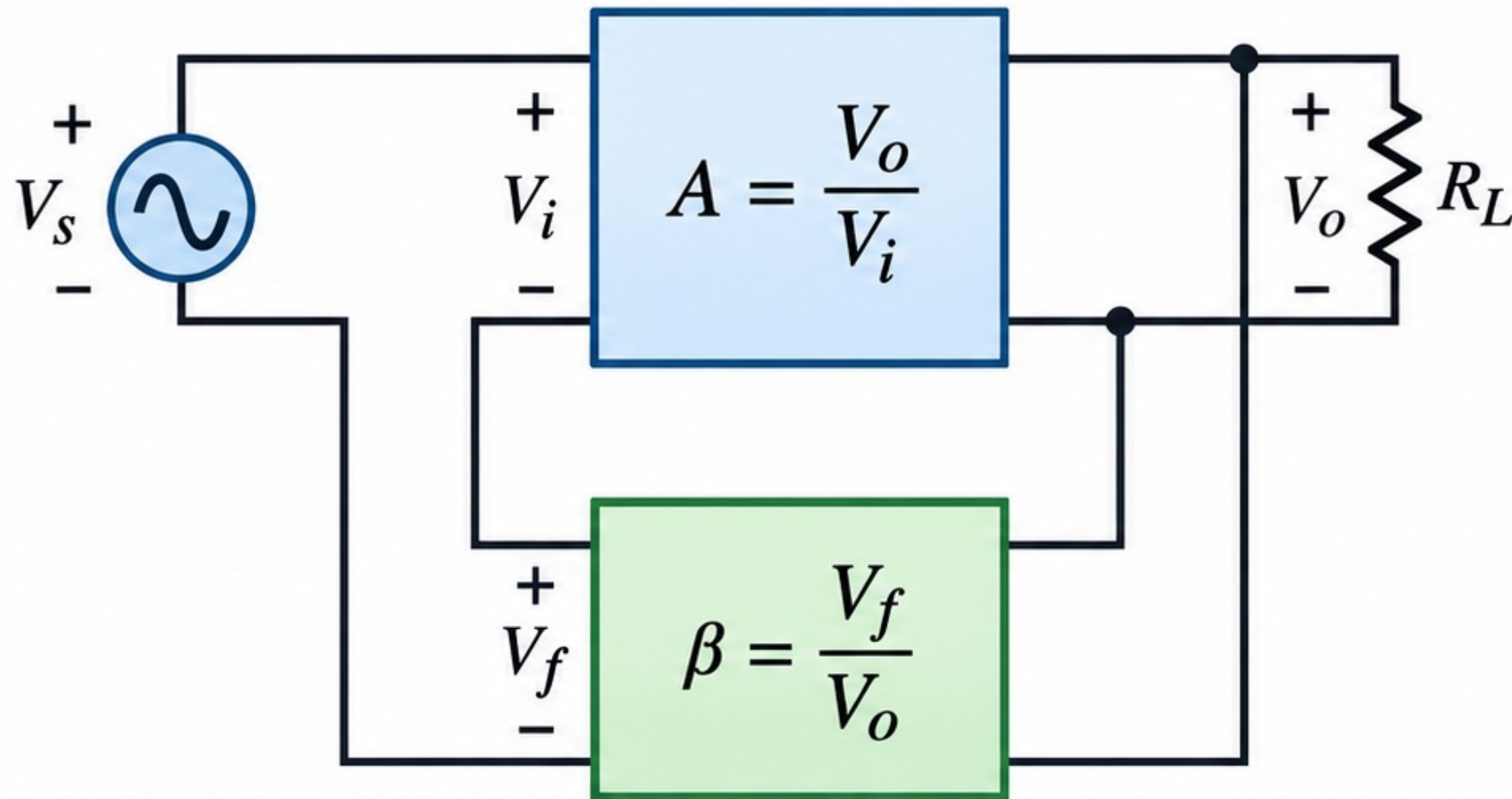


Feedback Amplifiers

Topologies, closed-loop gain, input/output impedances, and design insights

ANALYZE
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UNDERSTAND
APPLY

For Undergraduate Analog Electronics



Learning Goals

- 1 Understand why feedback is used
- 2 Identify the 4 basic feedback topologies
- 3 Derive closed-loop gain and impedance relations
- 4 Draw practical design inferences



Use the notation of the feedback block diagram throughout the deck.

2. Universal Closed-Loop Derivation

Deriving the standard negative-feedback closed-loop gain using A and β

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1 Input relation
(subtract feedback)

$$V_i = V_s - V_f$$

2 Amplifier relation

$$V_o = A V_i$$

3 Feedback relation

$$V_f = \beta V_o$$

4 Substitute
(use 1, 3 in 2)

$$V_o = A (V_s - \beta V_o)$$

5 Rearrange
(solve for V_o/V_s)

$$\frac{V_o}{V_s} = \frac{A}{1 + A\beta}$$

Loop gain

$$T = A\beta$$



Key Takeaways

- 1** Closed-loop gain is the open-loop gain divided by $1 + \text{loop gain}$.
- 2** If $T \gg 1$, then $\frac{V_o}{V_s} \approx \frac{1}{\beta}$.
- 3** Negative feedback reduces sensitivity to variations in A .
- 4** Sensitivity of closed-loop gain to A scales as $1/(1 + T)$.

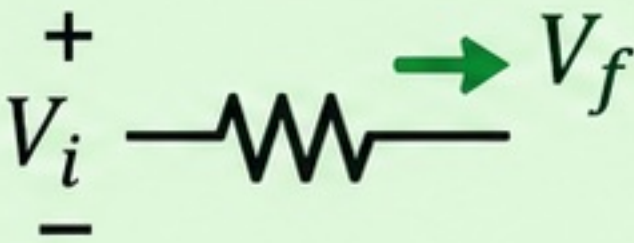
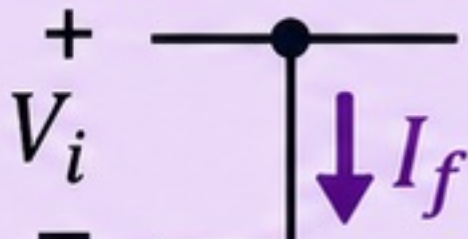


Takeaway: The denominator $1 + A\beta$ is the signature of negative feedback.

3. How to Identify the 4 Feedback Topologies

Classify by what is sensed at the output and what is returned at the input

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Quantity sensed at output	Signal returned at input	
	Voltage (series mixing)	Current (shunt mixing)
Voltage (shunt sensing) 	Voltage – Voltage (Series – Shunt) 	Voltage – Current (Shunt – Shunt) 
Current (series sensing) 	Current – Voltage (Series – Series) 	Current – Current (Shunt – Series) 



Key Rules

- 1** Series mixing at input $\Rightarrow$ input impedance **increases**.
 - 2** Shunt mixing at input $\Rightarrow$ input impedance **decreases**.
 - 3** Voltage sensing at output $\Rightarrow$ output impedance **decreases**.
 - 4** Current sensing at output $\Rightarrow$ output impedance **increases**.
-  The first word names the output quantity being sensed; the second word names the input quantity being returned.

3a. How to Derive $R_{in,f}$ and $R_{out,f}$

The test-source method: zero the input, apply V_x , and find V_x/I_x

The Recipe

- 1 Model the forward amplifier with a finite R_{in} and R_{out} , and keep the feedback network **ideal** (pure β , no loading).
- 2 **Zero the independent source:** $V_s = 0$ for series mixing, $I_i = 0$ (open) for shunt mixing.
- 3 Apply a test source V_x at the port of interest and find the current I_x it delivers.
- 4 Express the error signal (V_i or I_e) in terms of V_x , then propagate it through A and β .
- 5 Write **KVL** for a series connection or **KCL** for a shunt connection at that port.
- 6 Form $R = V_x/I_x$. The returned signal always *opposes* the test source, so the factor $1 + T$ appears.

Loop gain $T = \beta A_v, \beta R_m, \beta G_m, \beta A_i$: β times the relevant open-loop gain.

Where the Test Source Goes

- 1 **Series mixing (input).** V_x sits in series with R_{in} , so the same I_x flows through it: $R_{in,f} = R_{in}(1 + T)$.
- 2 **Shunt mixing (input).** V_x sits across the input node and I_x splits into I_e and I_f : $R_{in,f} = R_{in}/(1 + T)$.
- 3 **Voltage sensing (output).** V_x is applied in parallel with the output: $R_{out,f} = R_{out}/(1 + T)$.
- 4 **Current sensing (output).** V_x is inserted *in series* with the output wire: $R_{out,f} = R_{out}(1 + T)$.

Common mistake

When the loop senses *current*, never place V_x in parallel with the output: the feedback network must carry the same current as the test source, otherwise the answer comes out inverted.

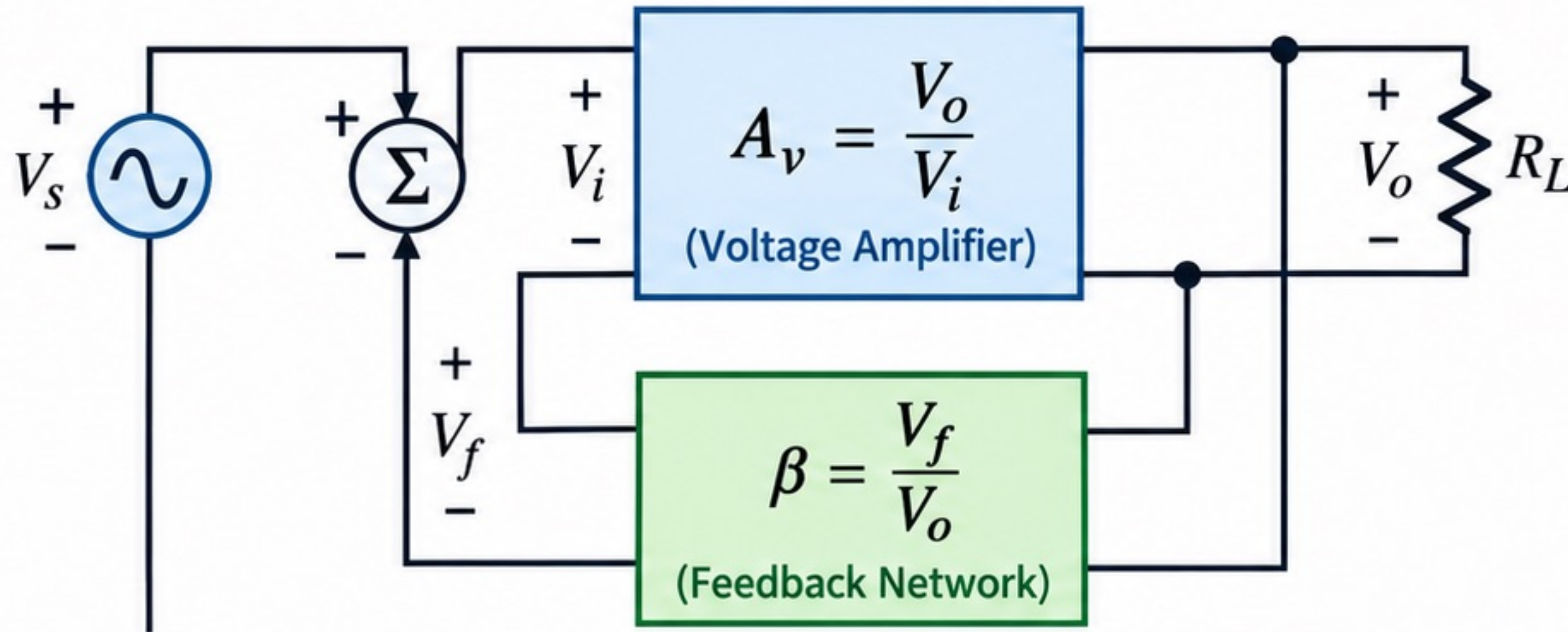
Feedback regulates whatever it senses and opposes whatever is mixed in, which is why every impedance moves by exactly $1 + T$.



4. Voltage-Voltage Feedback (Series-Shunt)

A voltage amplifier with negative feedback that samples the output voltage (shunt) and returns a voltage in series at the input.

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UNDERSTAND
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Derivation of Closed-Loop Gain

$$V_i = V_s - V_f \quad (1)$$

$$V_o = A_v V_i \quad (2)$$

$$V_f = \beta V_o \quad (3)$$

Substitute (2) and (3) into (1):

$$V_i = V_s - \beta V_o = V_s - \beta A_v V_i$$

$$V_i(1 + \beta A_v) = V_s$$

Closed-loop gain:

$$A_{v,f} = \frac{V_o}{V_s} = \frac{A_v}{1 + \beta A_v}$$

$$\text{Loop gain: } T = \beta A_v$$

Input and Output Impedances

$$R_{in,f} = R_{in}(1 + \beta A_v)$$

$$R_{out,f} = \frac{R_{out}}{1 + \beta A_v}$$

Key Inferences

- 1 Gain is reduced, but becomes more stable.
- 2 Input impedance rises.
- 3 Output impedance falls.
- 4 If $\beta A_v \gg 1$, then $A_{v,f} \approx 1/\beta$.
- 5 This is the preferred topology for a good voltage amplifier.



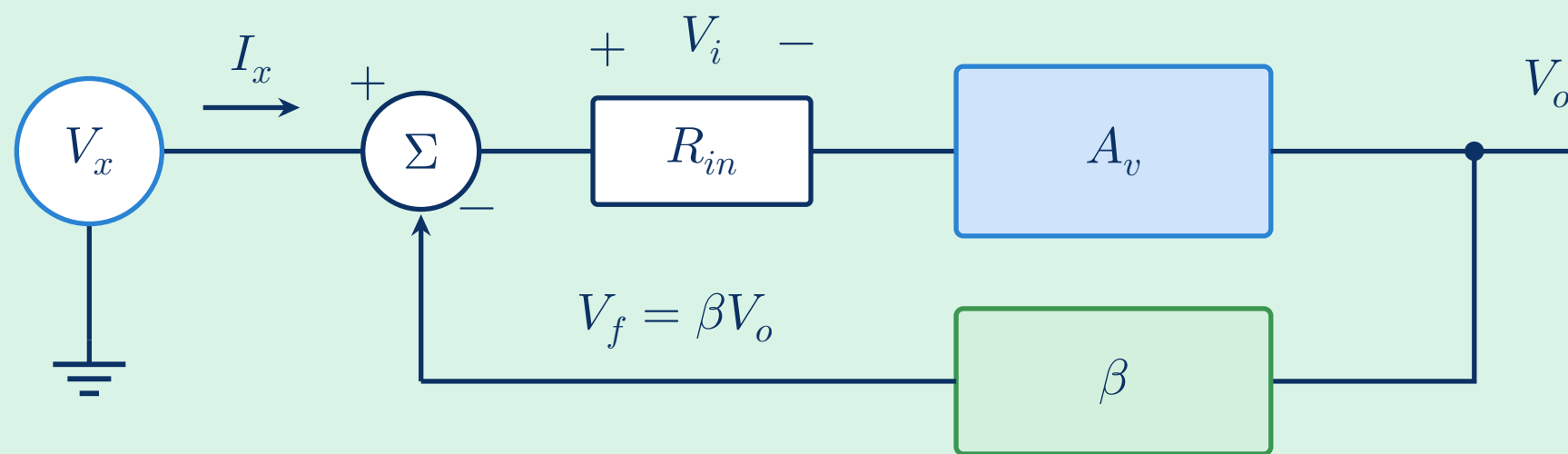
Use the notation of the feedback block diagram throughout the deck.

4a. Voltage–Voltage: Impedance Derivations

Series mixing raises R_{in} and voltage sensing lowers R_{out} , both by $1 + \beta A_v$

ANALYZE
DESIGN
UNDERSTAND
APPLY

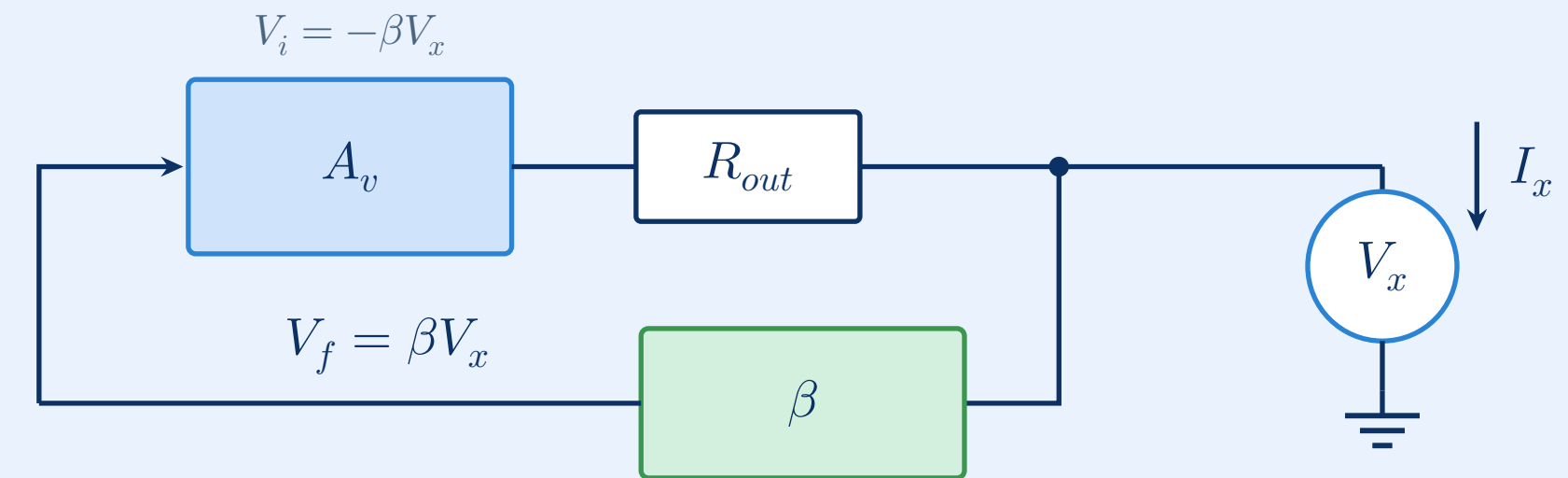
Input Resistance — Series Mixing



- 1 Series mixing puts R_{in} in the test loop, so $V_i = I_x R_{in}$.
- 2 The returned voltage is $V_f = \beta V_o = \beta A_v V_i = \beta A_v I_x R_{in}$.
- 3 KVL around the input loop: $V_x = V_i + V_f = I_x R_{in} (1 + \beta A_v)$.

$$R_{in,f} = \frac{V_x}{I_x} = R_{in} (1 + \beta A_v) = R_{in} (1 + T)$$

Output Resistance — Voltage Sensing



- 1 Set $V_s = 0$ and apply V_x across the output. The loop senses it, so $V_f = \beta V_x$ and $V_i = -V_f = -\beta V_x$.
- 2 The internal source of the forward amplifier becomes $A_v V_i = -\beta A_v V_x$.
- 3 Through R_{out} : $I_x = [V_x - (-\beta A_v V_x)] / R_{out} = V_x (1 + \beta A_v) / R_{out}$.

$$R_{out,f} = \frac{V_x}{I_x} = \frac{R_{out}}{1 + \beta A_v} = \frac{R_{out}}{1 + T}$$

The series–shunt loop pushes the stage toward an ideal voltage amplifier: high R_{in} , low R_{out} .



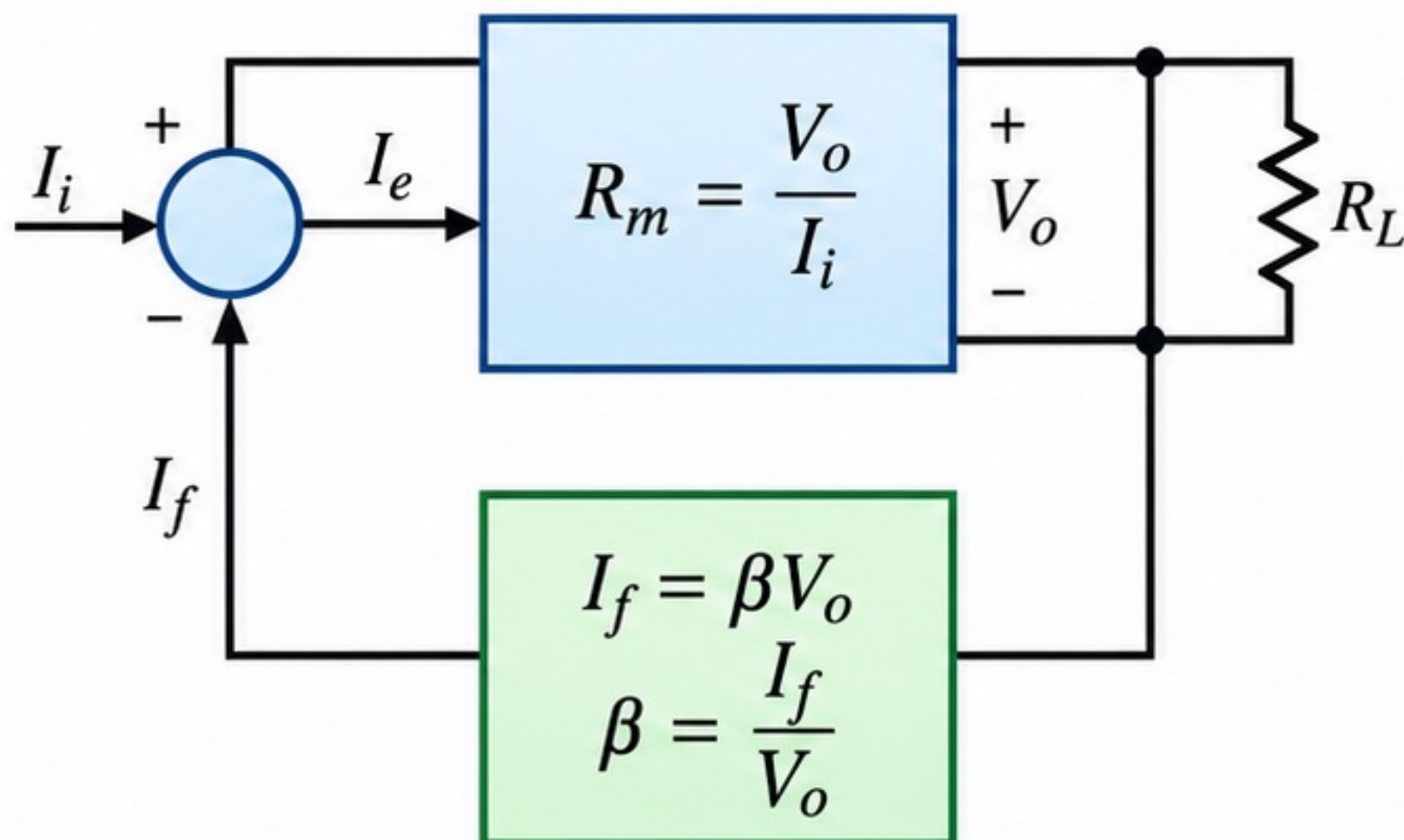
Use the notation of the feedback block diagram throughout the deck.

5. Voltage–Current Feedback (Shunt–Shunt)

Transresistance (Transimpedance) Configuration

ANALYZE
DESIGN
UNDERSTAND
APPLY

Block Diagram



Feedback network senses output voltage and returns a current to the input (*shunt–shunt*).

Key Equations

- 1 $I_e = I_i - I_f$
- 2 $V_o = R_m I_e$
- 3 $I_f = \beta V_o$

Therefore,

$$R_{m,f} = \frac{V_o}{I_i} = \frac{R_m}{1 + \beta R_m}$$

Loop gain:

$$T = \beta R_m$$

Input and Output Impedances

$$R_{in,f} = \frac{R_{in}}{1 + \beta R_m}$$
$$R_{out,f} = \frac{R_{out}}{1 + \beta R_m}$$

Design Insights

- 1 The feedback mixes current in shunt at the input, so R_{in} falls.
- 2 The feedback senses output voltage, so R_{out} falls.
- 3 β has units of conductance (siemens).
- 4 If $\beta R_m \gg 1$, the closed-loop transresistance is set mainly by $1/\beta$.



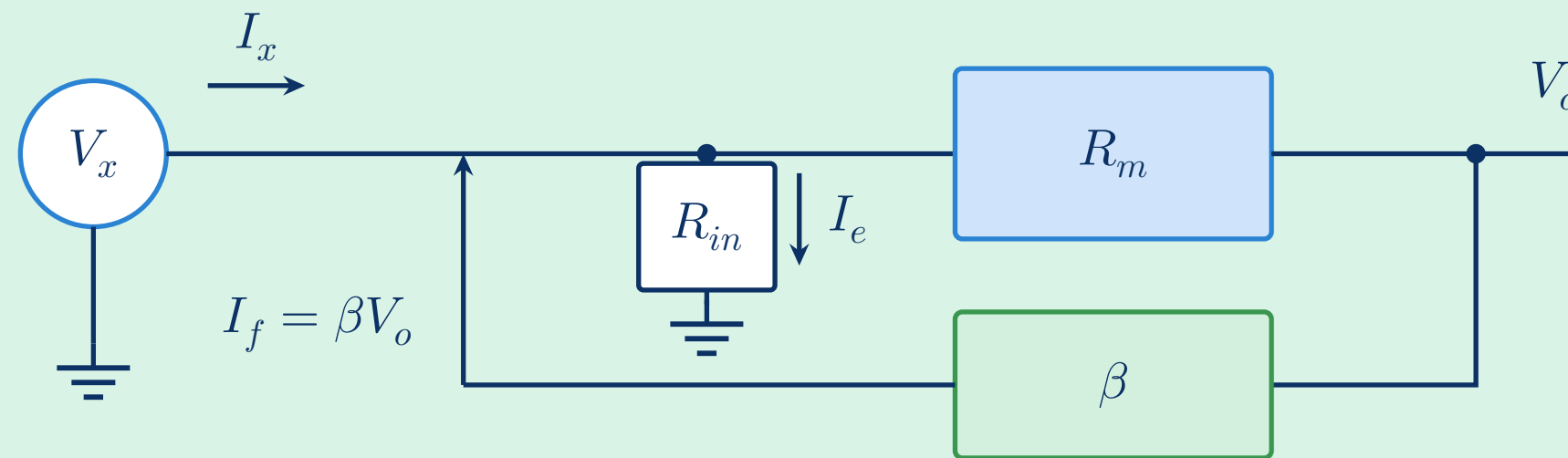
Use the notation of the feedback block diagram throughout the deck.

5a. Voltage–Current: Impedance Derivations

Shunt mixing lowers R_{in} and voltage sensing lowers R_{out} , both by $1 + \beta R_m$

ANALYZE
DESIGN
UNDERSTAND
APPLY

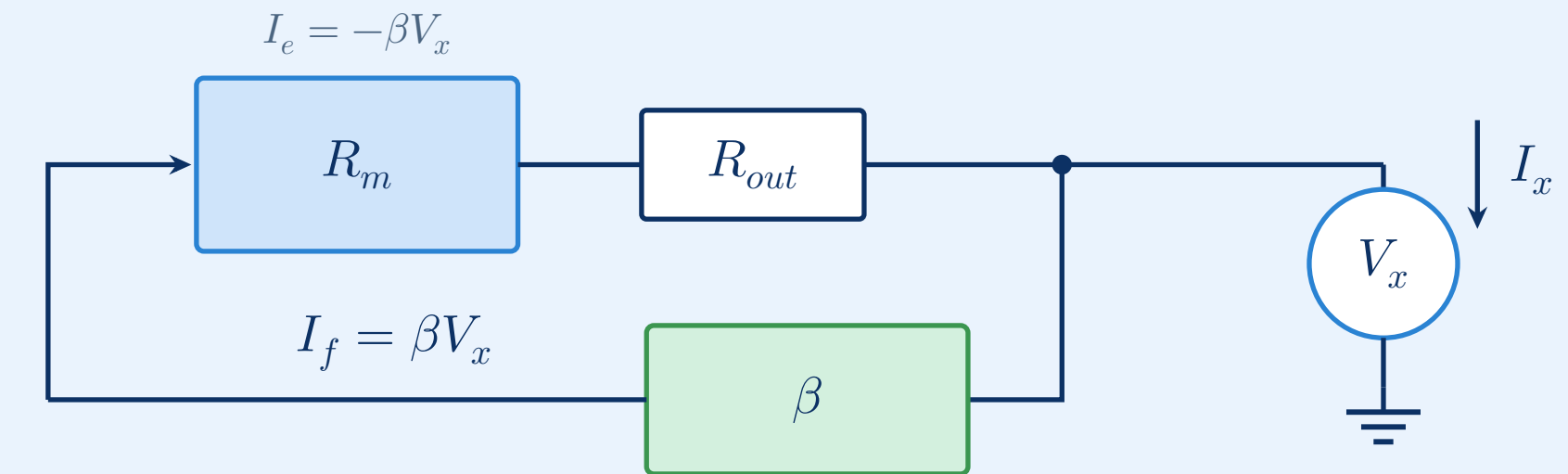
Input Resistance — Shunt Mixing



- 1 Apply V_x at the input node. With R_{in} in shunt, the error current is $I_e = V_x / R_{in}$.
- 2 Then $V_o = R_m I_e$, so the returned current is $I_f = \beta V_o = \beta R_m V_x / R_{in}$.
- 3 KCL at the input node: $I_x = I_e + I_f = (V_x / R_{in})(1 + \beta R_m)$.

$$R_{in,f} = \frac{V_x}{I_x} = \frac{R_{in}}{1 + \beta R_m} = \frac{R_{in}}{1 + T}$$

Output Resistance — Voltage Sensing



- 1 Open the input source ($I_i = 0$) and apply V_x at the output; the loop returns $I_f = \beta V_x$.
- 2 With the input open, $I_e = -I_f = -\beta V_x$, so the internal output source is $R_m I_e = -\beta R_m V_x$.
- 3 Through R_{out} : $I_x = [V_x - (-\beta R_m V_x)] / R_{out} = V_x(1 + \beta R_m) / R_{out}$.

$$R_{out,f} = \frac{V_x}{I_x} = \frac{R_{out}}{1 + \beta R_m} = \frac{R_{out}}{1 + T}$$

The shunt–shunt loop makes a good current sensor and a good voltage source: both port impedances fall by $1 + T$.

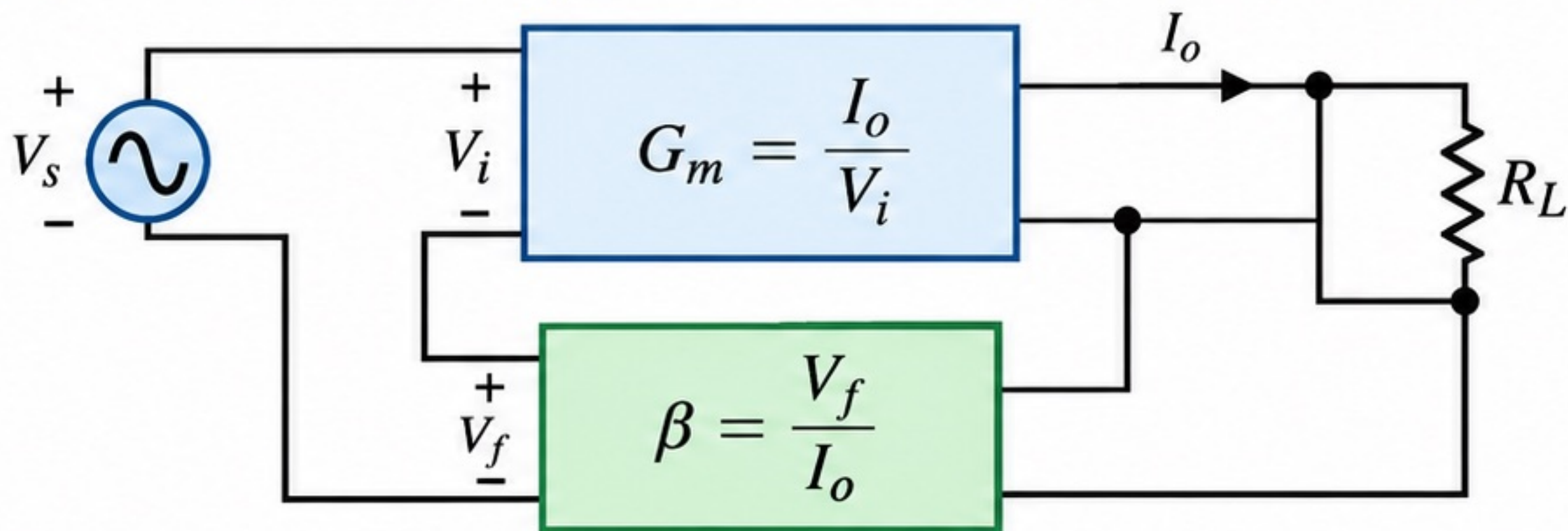


Use the notation of the feedback block diagram throughout the deck.

6. Current-Voltage Feedback (Series-Series)

Transconductance amplifier with current feedback

ANALYZE
DESIGN
UNDERSTAND
APPLY



Derivation

$$I_o = G_m (V_s - V_f)$$

$$V_f = \beta I_o$$

Therefore,

$$G_{m,f} = \frac{I_o}{V_s} = \frac{G_m}{1 + \beta G_m}$$

$$\text{Loop gain: } T = \beta G_m$$

Input and Output Impedances

$$R_{in,f} = R_{in} (1 + \beta G_m)$$

$$R_{out,f} = R_{out} (1 + \beta G_m)$$

Key Inferences

- 1 Series mixing at the input makes R_{in} increase.
- 2 Current sensing at the output makes R_{out} increase.
- 3 β has units of resistance (ohms).
- 4 If $\beta G_m \gg 1$, then $I_o / V_s \approx 1/\beta$.
- 5 This topology is useful when we want a more ideal current source output.

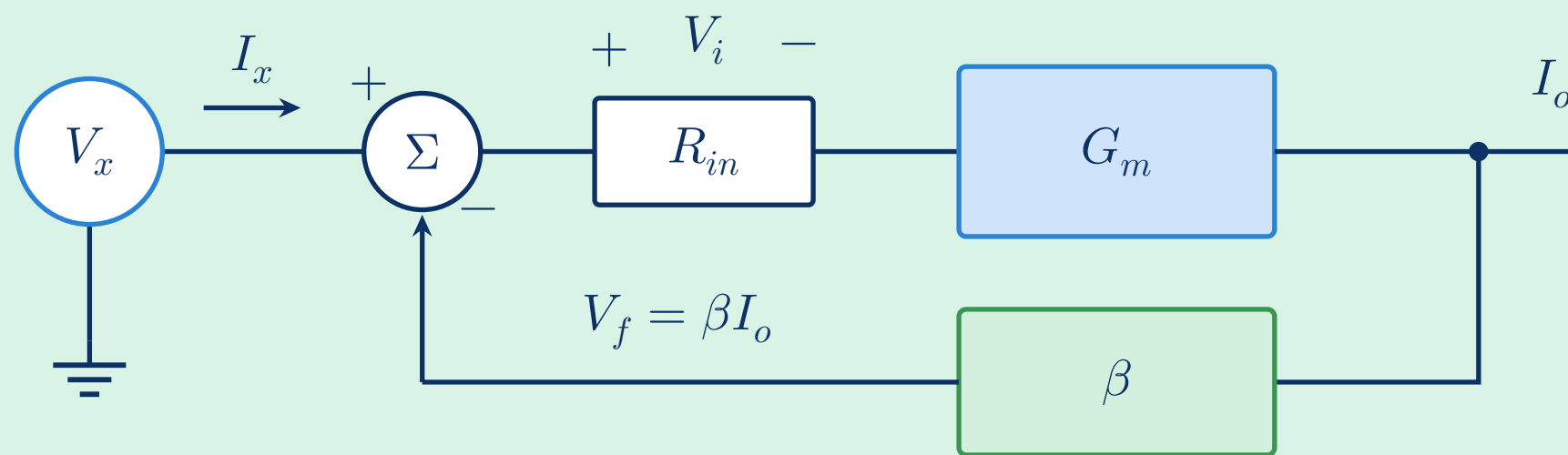


Use the notation of the feedback block diagram throughout the deck.

6a. Current-Voltage: Impedance Derivations

Series mixing raises R_{in} and current sensing raises R_{out} , both by $1 + \beta G_m$

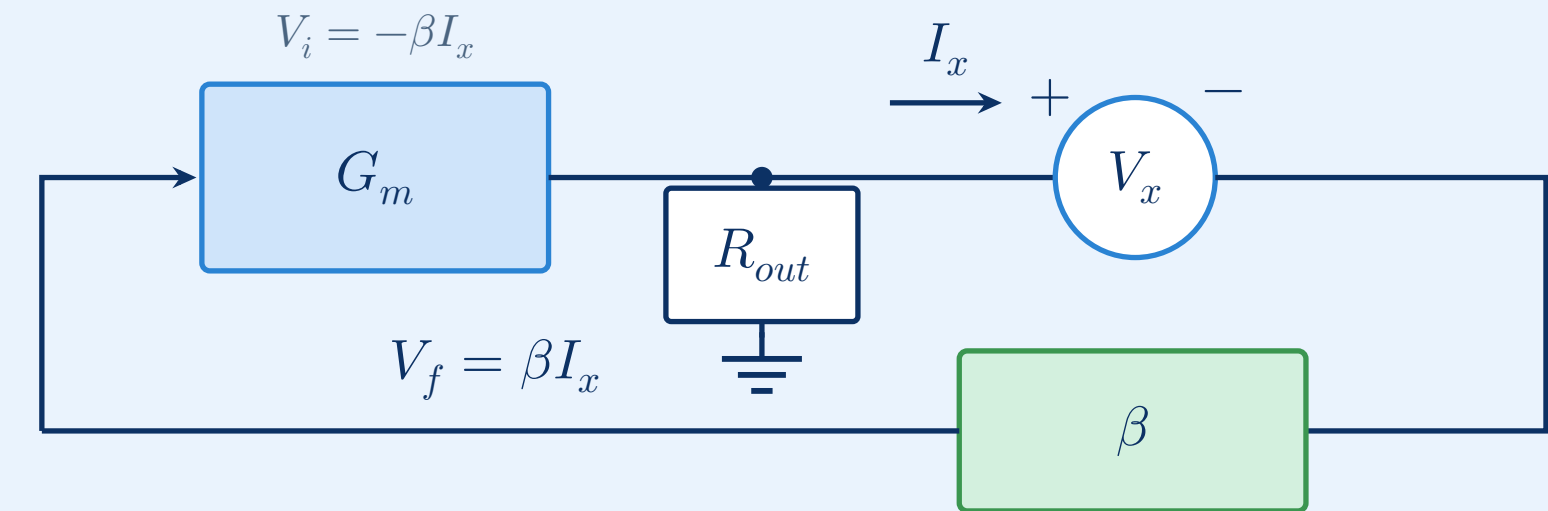
Input Resistance — Series Mixing



- 1 The test current flows through R_{in} , so $V_i = I_x R_{in}$.
- 2 The loop senses the output current: $V_f = \beta I_o = \beta G_m V_i = \beta G_m I_x R_{in}$.
- 3 KVL around the input loop: $V_x = V_i + V_f = I_x R_{in} (1 + \beta G_m)$.

$$R_{in,f} = \frac{V_x}{I_x} = R_{in} (1 + \beta G_m) = R_{in} (1 + T)$$

Output Resistance — Current Sensing



- 1 Insert V_x in series with the output wire; the feedback network then carries the same current, so $V_f = \beta I_x$.
- 2 Hence $V_i = -\beta I_x$, and the forward stage sources $G_m V_i = -\beta G_m I_x$ into the output node.
- 3 KCL at the output node: $I_x = V_x / R_{out} - \beta G_m I_x$.

$$R_{out,f} = \frac{V_x}{I_x} = R_{out} (1 + \beta G_m) = R_{out} (1 + T)$$

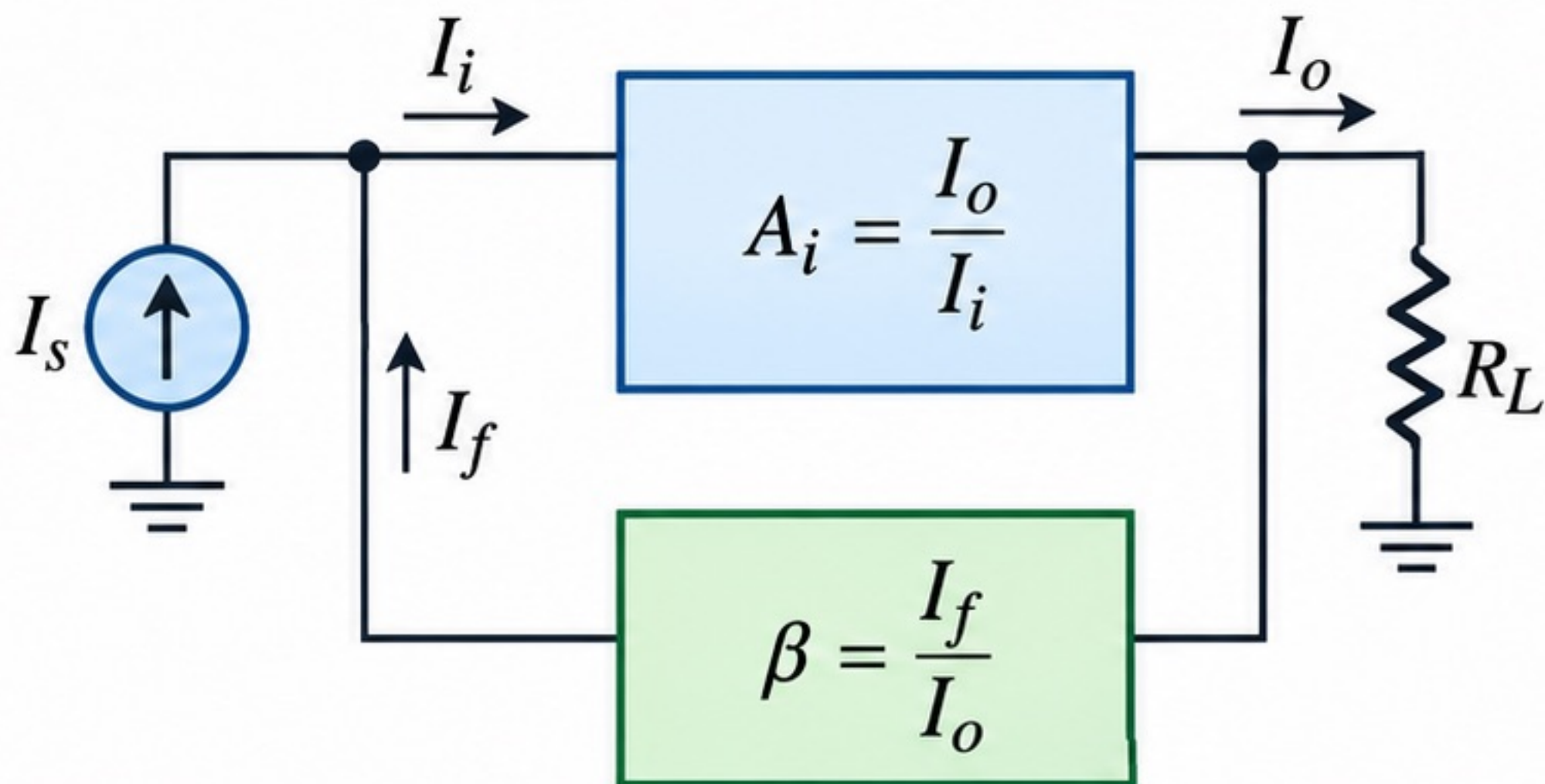
The series-series loop pushes the stage toward an ideal transconductor: a stiff current output with a high R_{out} .



7. Current-Current Feedback (Shunt-Series)

Current amplifier: current sensing at output, current mixing at input

ANALYZE
DESIGN
UNDERSTAND
APPLY



Input/Output Impedances

Input Impedance
(shunt mixing)

$$R_{in,f} = \frac{R_{in}}{1 + \beta A_i}$$

Output Impedance
(current sensing)

$$R_{out,f} = R_{out}(1 + \beta A_i)$$

Derivation

$$I_o = A_i (I_i - I_f)$$

$$I_f = \beta I_o$$

$$A_{i,f} = \frac{I_o}{I_i} = \frac{A_i}{1 + \beta A_i}$$

$$\text{Loop gain: } T = \beta A_i$$



Learning Takeaways

1

Shunt mixing at the input lowers R_{in} .

2

Current sensing at the output raises R_{out} .

3

β is dimensionless in the current amplifier case (since it is a current ratio).

4

If $\beta A_i \gg 1$, the closed-loop gain becomes much less sensitive to variations in A_i .

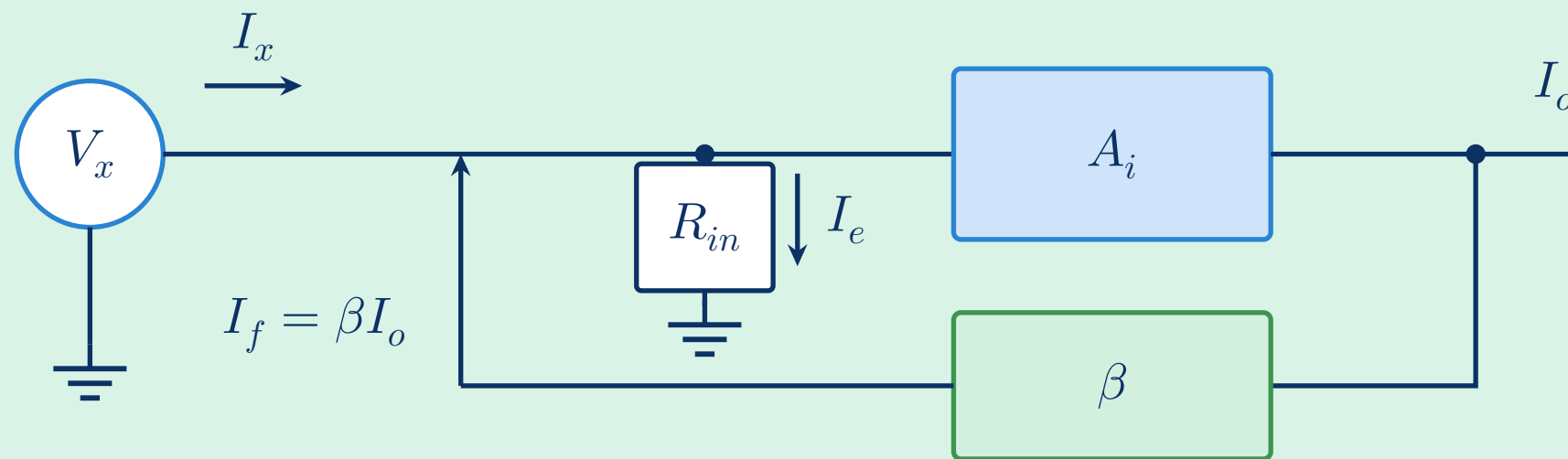


Use the notation of the feedback block diagram throughout the deck.

7a. Current–Current: Impedance Derivations

Shunt mixing lowers R_{in} and current sensing raises R_{out} , both by $1 + \beta A_i$

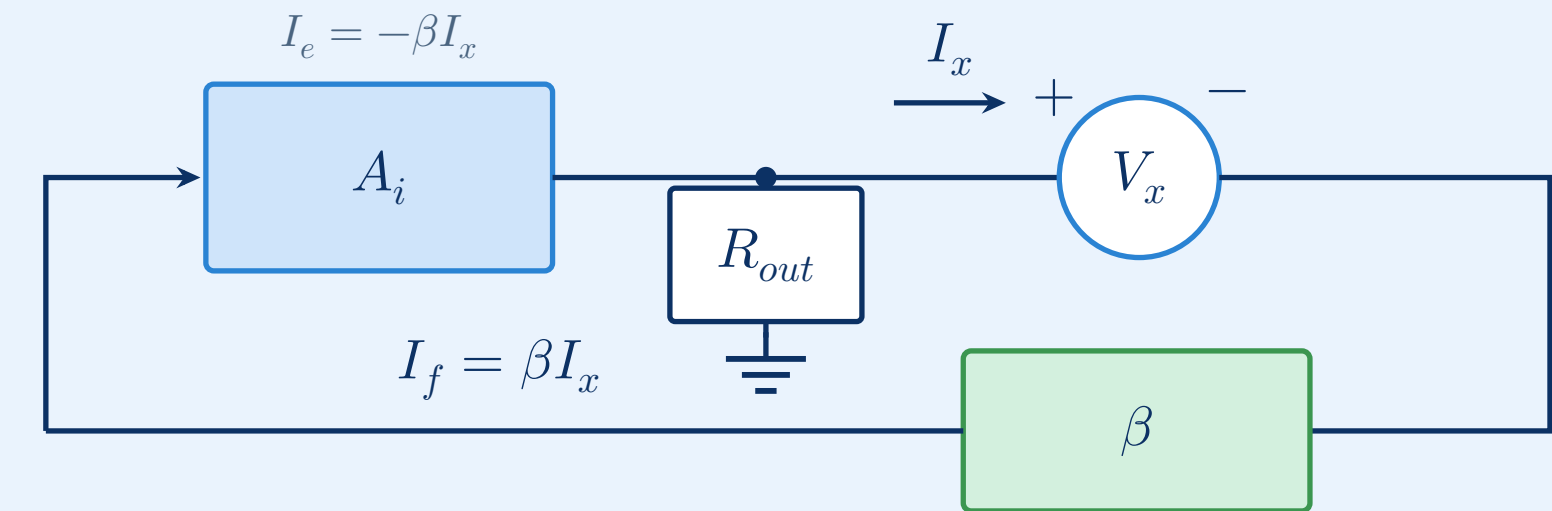
Input Resistance — Shunt Mixing



- 1 Apply V_x at the input node; the error current through R_{in} is $I_e = V_x/R_{in}$.
- 2 Then $I_o = A_i I_e$, so the returned current is $I_f = \beta I_o = \beta A_i V_x / R_{in}$.
- 3 KCL at the input node: $I_x = I_e + I_f = (V_x/R_{in})(1 + \beta A_i)$.

$$R_{in,f} = \frac{V_x}{I_x} = \frac{R_{in}}{1 + \beta A_i} = \frac{R_{in}}{1 + T}$$

Output Resistance — Current Sensing



- 1 Leave the input open ($I_i = 0$) and insert V_x in series with the output, so $I_f = \beta I_x$.
- 2 With the input open, $I_e = -I_f = -\beta I_x$, and the forward amplifier delivers $A_i I_e = -\beta A_i I_x$.
- 3 KCL at the output node: $I_x = V_x/R_{out} - \beta A_i I_x$.

$$R_{out,f} = \frac{V_x}{I_x} = R_{out} (1 + \beta A_i) = R_{out} (1 + T)$$

The shunt–series loop makes the stage a better current amplifier: it accepts current easily and delivers it stiffly.



8. Universal Gain and Impedance Rules

One-slide summary of all four negative-feedback topologies

ANALYZE
DESIGN
UNDERSTAND
APPLY

Topology	Forward gain	Closed-loop gain	Effect on R_{in}	Effect on R_{out}
1 Voltage–Voltage (Series–Shunt)	$A_v = \frac{V_o}{V_i}$	$A_{v,f} = \frac{A_v}{1 + \beta A_v}$	increases by $(1 + \beta A_v)$	decreases by $(1 + \beta A_v)$
2 Voltage–Current (Shunt–Shunt)	$R_m = \frac{V_o}{I_i}$	$R_{m,f} = \frac{R_m}{1 + \beta R_m}$	decreases by $(1 + \beta R_m)$	decreases by $(1 + \beta R_m)$
3 Current–Voltage (Series–Series)	$G_m = \frac{I_o}{V_i}$	$G_{m,f} = \frac{G_m}{1 + \beta G_m}$	increases by $(1 + \beta G_m)$	increases by $(1 + \beta G_m)$
4 Current–Current (Shunt–Series)	$A_i = \frac{I_o}{I_i}$	$A_{i,f} = \frac{A_i}{1 + \beta A_i}$	decreases by $(1 + \beta A_i)$	increases by $(1 + \beta A_i)$



Input rule:

series mixing raises R_{in} ; shunt mixing lowers R_{in} .



Output rule:

voltage sensing lowers R_{out} ; current sensing raises R_{out} .

| Feedback pushes the amplifier toward its ideal role.

9. Analysis Checklist and Design Insights

A step-by-step guide to analyzing and designing feedback amplifiers

ANALYZE
DESIGN
UNDERSTAND
APPLY

- 1 Identify the forward amplifier type: A_v , R_m , G_m , or A_i .
- 2 Identify what the feedback network **senses** at the output.
- 3 Identify what it **returns** to the input.
- 4 Confirm that the feedback is **negative**: the returned signal must oppose the original change.
- 5 Compute loop gain $T = A\beta$ (or β times the appropriate open-loop gain).
- 6 Write the closed-loop gain as $A_{CL} = \frac{A}{1 + T}$
- 7 Use the **topology rules** to find R_{in} and R_{out} .
- 8 **Check assumptions**: ideal feedback network, loading effects, and sufficiently large loop gain.



Design Insights



Feedback trades gain for **precision** and **robustness**.



It **reduces sensitivity** to device/model parameter variations.



It can make a voltage amplifier **more ideal** or a current source **more ideal**, depending on topology.



Real circuits must also consider **loading and stability**.



Master the topology first; the formulas then become almost automatic.